

Review 3

Helicopter rotors are designed to be flexible in flapping, typically about a mechanical hinge. This has been standard practice since Cierva added a flapping hinge to his gyrocopter rotors to address cyclical loading issues encountered in forward flight that had led to the structural failure of rotors during early attempts at vertical flight. For a rotor blade of radius R , rotating at an angular velocity Ω , the flapping equations of motion about the flapping hinge is

$$I_b \frac{d^2 \beta}{dt^2} + I_b \Omega^2 \cos \beta \sin \beta = \int_0^R \frac{1}{2} \rho V_r^2 c C_l y dy$$

where β is the flapping angle and $V_r(y)$ and C_L are the apparent velocity and lift coefficient at a blade section of chord c at a distance y outboard of the hinge.

- (i) If the rotorcraft of mass m is in equilibrium, derive expressions for the equilibrium conditions of its flapping states. You may assume the rotor in question has $N_b = 2$ blades.
- (ii) Pick an appropriate set of states and express the dynamics of the system in state-space form.
- (iii) Use your state space model to characterise the dynamic response of the system to perturbations in its flap angle.
- (iv) Modern rotorcraft have eliminated mechanical hinges in favour of flexible structural members, typically manufactured using composite materials. If the stiffness of the blade at the location of the hinge is now K_β , plot a root-locus diagram illustrating the effect of K_β on the system's dynamics.

(a) Plain flapping hinge. (b) Delta-three hinge combines flapping and cyclic feathering.

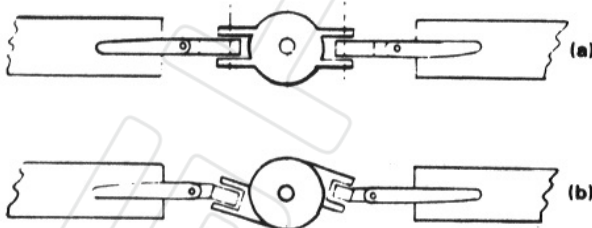
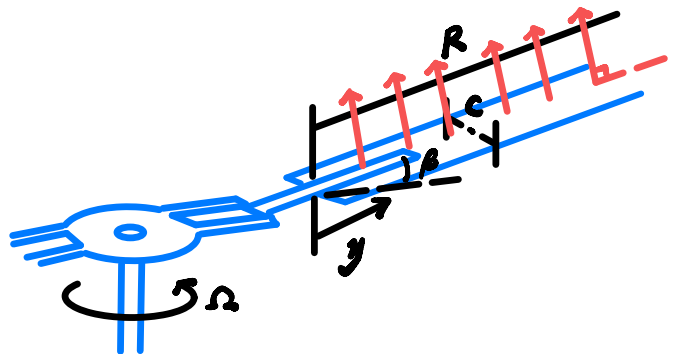


FIGURE 2-51. FLAPPING HINGES.



Assume $C_L(y)$ and $c(y)$ constant

$$(i) \underbrace{I_b \ddot{\beta}}_{\text{inertia}} + \underbrace{I_b \Omega^2 \cos \beta \sin \beta}_{\text{centripital}} = \underbrace{\int_0^R \frac{1}{2} \rho V_r^2 c C_L y dy}_{\text{lift} \cdot y} \quad * V_r = \Omega \cos \beta \cdot y$$

$$x = [\beta, \beta]$$

$$\therefore \ddot{x} = [\ddot{\beta}, \ddot{\beta}]$$

At equilibrium, $\ddot{x} = 0$

$$\begin{aligned} I_b \Omega^2 \sin \beta_e \cos \beta_e &= \int_0^R \frac{1}{2} \rho \Omega^2 \cos^2 \beta_e (C_L)_e y^2 dy \\ &= \frac{1}{8} \rho \Omega^2 \cos^2 \beta_e (C_L)_e R^4 \end{aligned}$$

$$\begin{aligned} \text{Also, } mg = F_z &= N_b \int_0^R \frac{1}{2} \rho V_r^2 c (C_L)_e dy \cdot \cos \beta \\ &= \frac{N_b}{6} \rho \Omega^2 \cos^3 \beta_e c (C_L)_e R^3 \\ &= \frac{1}{3} \rho \Omega^2 \cos^3 \beta_e c (C_L)_e R^3 \text{ for } N_b = 2 \end{aligned}$$

$$\therefore (C_L)_e = \frac{3mg}{\rho \Omega^2 \cos^3 \beta_e R^3 c}$$

$$\therefore I_b \Omega^2 \sin \beta_e \cos \beta_e = \frac{1}{8} \rho \Omega^2 \cos^2 \beta_e c \frac{3mg}{\rho \Omega^2 \cos^3 \beta_e R^3 c} R^4$$

$$\therefore \sin \beta_e \cos^2 \beta_e \approx \beta_e = \frac{3mgR}{8 I_b \Omega^2}$$

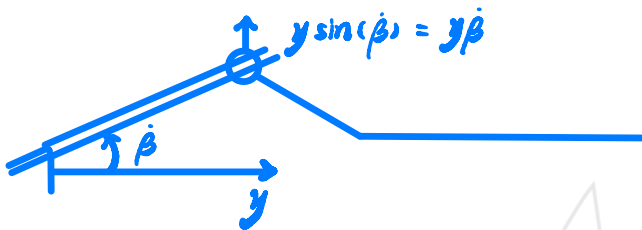
$$(ii) I_b \ddot{\beta} = -I_b \Omega^2 \cos \beta \sin \beta + M^a = -\frac{1}{2} I_b \Omega^2 \sin(2\beta) + M^a$$

$$\text{where } M^a = \int_0^R \frac{1}{2} \rho V_r^2 c C_{Ly} dy$$

$$\mathbf{x} = [\beta, \dot{\beta}] \quad \therefore \ddot{\mathbf{x}} = [\ddot{\beta}, \dot{\beta}]$$

Try to understand that $\dot{\beta}y$ is similar to W' ,

while β does not affect any α for cross section



$$V_r = \sqrt{(\Omega y \cos \beta)^2 + (\dot{\beta} y)^2} \quad * U_e = (V_r)_e = \Omega y \cos \beta$$

for $V_r(\beta, \dot{\beta})$

$$\therefore V_r^2 = (\Omega y \cos \beta)^2 + \dot{\beta}^2 y^2$$

$$\therefore \frac{dV_r^2}{d\beta} = \frac{d}{d\beta} \Omega^2 y^2 \left[\frac{1}{2} + \frac{1}{2} \cos(2\beta) \right] = -\Omega^2 y^2 \sin(2\beta)$$

$$= -2 \Omega^2 y^2 \sin \beta \cos \beta = -2 \Omega^2 y^2 \beta$$

$$\therefore \beta_e = \frac{3mgR}{8I_b \Omega^2} \quad \therefore \frac{dV_r^2}{d\beta} = -\frac{3mgR y^2}{4I_b}$$

$$\therefore \frac{dV_r^2}{d\beta} = 2\dot{\beta}y^2 = 0 \quad \text{---} \quad \dot{\beta}_e = 0$$

$$\text{for } C_L(\beta, \dot{\beta}), \quad \frac{C_L}{\dot{\alpha}_L} = a_L$$

$$\alpha_L = (\alpha_L)_e + \alpha' = (\alpha_L)_e - \tan^{-1}\left(\frac{\dot{\beta}r}{(Vr)_e}\right) = (\alpha_L)_e - \frac{\dot{\beta}y}{(Vr)_e}$$

$$\therefore \frac{d\alpha_L}{d\beta} = 0, \quad \frac{d\alpha_L}{d\dot{\beta}} = -\frac{y}{(Vr)_e} = -\frac{y}{\Omega \cos \beta y} = -\frac{1}{\Omega}$$

$$\therefore \frac{dC_L}{d\dot{\beta}} = \frac{dC_L}{d\alpha_L} \cdot \frac{d\alpha_L}{d\dot{\beta}} = -\frac{a_L}{\Omega}$$

$$\therefore \frac{dM^a}{d\beta} = \frac{dM^a}{dVr^2} \frac{dVr^2}{d\beta} + \frac{dM^a}{dC_L} \frac{dC_L}{d\dot{\beta}} = \frac{dM^a}{dVr^2} \frac{dVr^2}{d\beta} + 0$$

$$= \int_0^R \frac{1}{2} \rho C_L y dy \cdot -\frac{3mgRy^2}{4I_b}$$

$$= \int_0^R \frac{3\rho cmgR C_L y^3}{8I_b} dy = -\frac{3\rho cmgR^4 C_L}{32I_b}$$

$$\therefore (C_L)_e = \frac{3mg}{\rho \Omega^2 \cos^3 \beta_e R^3 c}$$

$$\therefore M_\beta^a = \frac{3\rho cmgR^4}{32I_b} \frac{3mg}{\rho \Omega^2 \cos^3 \beta_e R^3 c} = -\frac{9m^2 g^2 R}{32I_b \Omega^2}$$

$$\therefore M_\beta^a = \frac{dM^a}{dVr^2} \frac{dVr^2}{d\beta} + \frac{dM^a}{dC_L} \frac{dC_L}{d\dot{\beta}} = 0 + \frac{dM^a}{dC_L} \frac{dC_L}{d\dot{\beta}}$$

$$= \int_0^R \frac{1}{2} \rho Vr^2 c y dy \frac{a_L}{\Omega}$$

$$= \int_0^R \frac{1}{2} \rho \Omega \cos^3 \beta a_L c y^3 dy$$

$$= \frac{1}{8} \rho \Omega \cos^3 \beta a_L c R^4 = \frac{1}{8} a_L \rho \Omega c R^4$$

$$\therefore \frac{d}{d\beta} \left[-\frac{1}{2} I_b \Omega^2 \sin(2\beta) \right] = -I_b \Omega^2 \cos(2\beta) = -I_b \Omega^2 [2 \cos\beta \cos\beta - 1] \\ = -I_b \Omega^2$$

$$\therefore \begin{bmatrix} I_b & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \ddot{\beta} \\ \dot{\beta} \end{bmatrix} = \begin{bmatrix} -\frac{a \rho n c R^4}{8} & -I_b \Omega^2 - \frac{3 \rho c m g R^4 C_b}{32 I_b} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{\beta} \\ \beta \end{bmatrix}$$

iii) iv) see solution...